Skin Disease Classifier

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Abstract

Skin diseases affect millions of people worldwide and can result in severe health complications if undetected or misdiagnosed. Early and accurate diagnosis is crucial in preventing the spread of further infection. This paper proposes a Convolutional Neural Network (CNN) approach for classifying skin diseases from images. We benchmark our CNN against a baseline logistic regression classifier as well as with a Vision Transformer as a state-of-the-art approach. Our goal is to achieve a classification accuracy of at least 90% as well as high F1 and AUPRC scores, with an emphasis on reducing false negatives. We train and evaluate our methods using a subset of images from the Dermnet dataset on Kaggle. Our model will provide dermatologists and healthcare professionals of varying experience levels with a reliable tool for diagnosing skin diseases and guiding patient treatment decisions. All work will be done on the following GitHub Repository: `https://github.com/manigel8/ECE324-Project.git`.

1 Introduction

Skin diseases are among the most common health issues, affecting millions of individuals worldwide. These diseases can range from relatively harmless, benign conditions such as moles, to more severe diseases such as melanoma. Early detection is crucial, as untreated skin diseases can spread or worsen, leading to increased risks and more complex treatment requirements.

However, identifying skin conditions accurately relies on visual inspection and knowledge from healthcare professionals. This can become challenging, especially for less experienced dermatologists, when given multiple possible diagnoses.

As a result, we propose a machine learning-based solution to improve the diagnostic process. By using convolutional neural networks (CNNs), we aim to provide accurate, automated classification of selected skin diseases.

1.1 Context and Motivation

In many areas of the health industry, there is a shortage of dermatologists, and patients often experience delays in receiving diagnoses. Even in large hospitals, healthcare professionals may face difficulties managing the overwhelming number of patients, which can result in the doctors becoming overworked. By automating part of the diagnostic process with a machine learning model, we not only increase the speed and efficiency in disease recognition but also minimize human error in interpreting skin images.

Additionally, from a public health standpoint, identifying skin diseases early can reduce healthcare costs in the long run. This is because in the early stages of skin diseases, treatment is often cheaper and more effective than attempts at treatment in the later stages when the skin disease becomes more prominent. Moreover, an automated system that performs at or above the level of an experienced dermatologist can significantly improve the quality of care for patients.

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1.2 Conditions of Success

We define our success criteria in terms of both **accuracy** and **patient safety**:

- **Accuracy** $\geq 90\%$**:** We aim for a classification accuracy that meets or exceeds the standard benchmark for medical diagnostic tasks [1]. Achieving or surpassing 90% accuracy is crucial for real-world clinical implementation.
- **Minimizing False Negatives and False Positives:**
  - **False Negatives (Diseases Missed):** Missing a serious skin disease can be extremely harmful because it may delay crucial treatment. The patient may not be able to receive the proper early treatment leading to less effective treatments later on. Therefore, the model should be especially sensitive to potential missed cases.
  - **False Positives (Overdiagnosis):** Although false positives are not as critical as false negatives, an excessive amount of false positives can lead to unnecessary anxiety in the patient and additional medical costs. Moreover, over-diagnosing leads to consequences of the unneeded follow-up treatments which can negatively affect patients psychologically and financially. However, given a choice, our bias slightly favors false positives, since a patient can always be examined further by a specialist.
- **F1 Score:** We use the F1 score to balance precision (avoiding false positives) and recall (capturing actual disease cases). A high F1 score indicates reliable overall performance.

These metrics directly influence the effectiveness of our approach in real health clinics. We prioritize patient safety and accuracy of the model, which aligns with a general principle in medical diagnostics: "Better to over-call the possibility of disease than to miss a genuine case" [2].

2 Data

2.1 Dataset Description

The dataset used for this project is the **Dermnet dataset** from Kaggle, which contains a large set of coloured, high-quality skin disease images. This dataset comprises of 20 different skin diseases, although not all diseases are equally represented. After conducting some initial data exploration, we discovered imbalances within the training and testing sets. Although our dataset provides us with a variety of training data from a diverse range of skin diseases, it was evident that some classes were far more robust than others and contained many more samples. For example, there were fewer than 100 data points for a severe disease such as melanoma and 800 data points for acne. Figure 1 shows the distribution of images within the dataset.

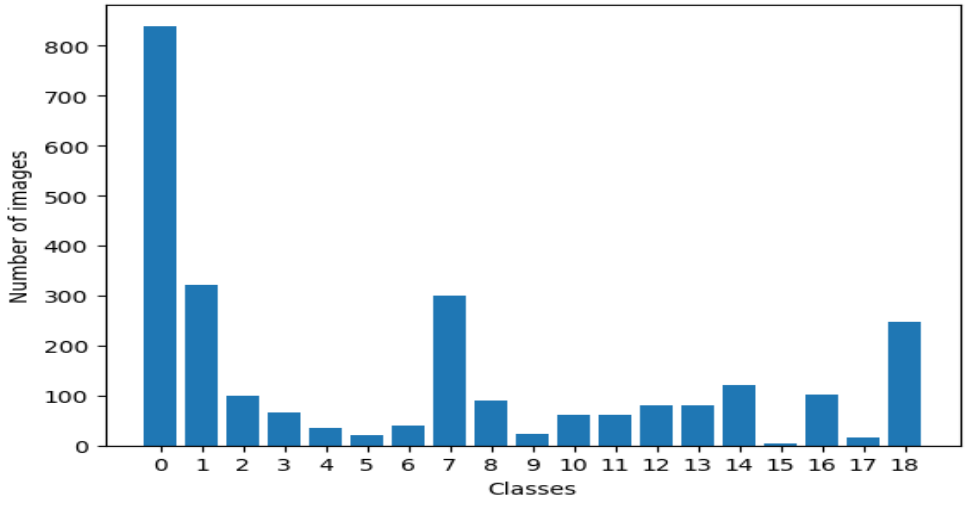


Figure 1: Data distribution in training data

The implications of this is significant as it means that there would be a tendency for any model we develop to overfit towards the dominant label, due to the abundance of training data is for it. This would lead to poor test accuracy and inaccurate classifications when evaluating the models on the testing set. The real-world implications may be significant, as more severe skin diseases that require earlier identification and treatment may be disregarded, leading to severe health consequences later in time.

2.2 Data Augmentation

To combat imbalances and enhance robustness, we employed two main techniques. The dataset was first scoped down, and any labels that contained too few training and test samples were removed, as they would only serve to add noise to our models. The scoping process consisted of evaluating how many samples there were for a given label, as well as how severe and prevalent the skin diseases were. In this way, the dataset still retains a set of skin diseases that range in both severity and rarity. Despite the scoping process, the dataset was still quite imbalanced, as the class "acne" still had more than twice the number of samples compared to every other label. Even if this dataset were used, there would still be overfitting and poor test accuracy. To resolve this issue, data augmentation was performed on the underrepresented labels as shown in Figure 2. The data augmentation techniques used include:

- **Random rotations and flips**: Due to the fact that image data is rotation invariant, we can create synthetic data by flipping images vertically and horizontally as well as adding rotations of various degrees (30 for the model). These noisy inputs will also make the models more robust, as they will be less sensitive to receiving images of only a certain orientation
- **Image resizing**: The images were also resized and normalized (e.g., to 224×224 pixels) to be consistent with common CNN and Vision Transformer input dimensions [3]

Data augmentation effectively increases the diversity of training examples, helping in generalization, especially for underrepresented classes where the number of samples is small.

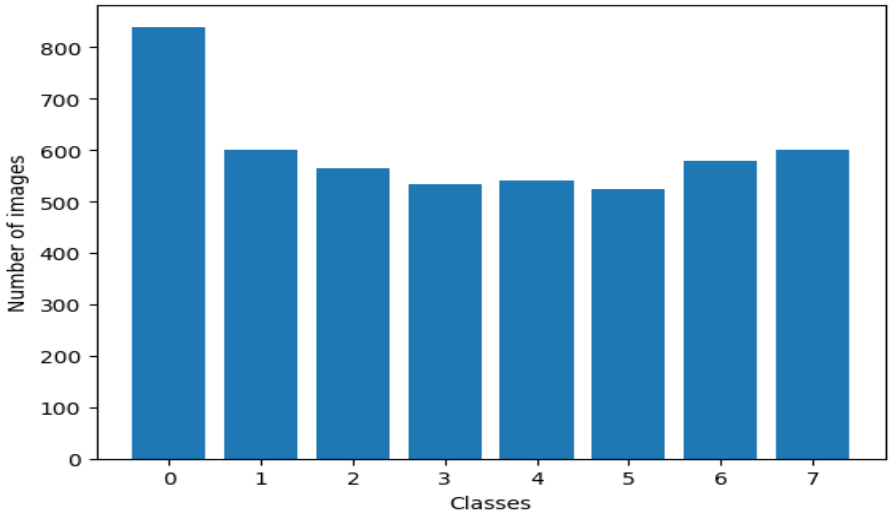


Figure 2: Data distribution in balanced, synthetic training data

2.3 Input and Output

- **Input:** Cropped and color images of skin or patches from the Dermnet dataset. We will standardize image sizes to fit the input layer requirements of our CNN.
- **Output:** The model outputs a label of which disease a patient potentially has based on the image inputted along with a probability of how certain the model is.

Our approach aligns with the reality of clinical treatments where a healthcare professional might first confirm whether a patient has an abnormality or not. If a disease is confirmed, they would then identify which disease it corresponds to and provide the proper treatment.

3 Modeling Approach

3.1 Choice of Model: Convolutional Neural Network (CNN)

A **CNN** is suitable for analyzing image data because it can autonomously learn hierarchical feature representations. It begins by detecting edges and simple shapes in early layers to more complex textures and patterns in deeper layers. Dermatological diagnosis commonly depends on details such as color, shape, texture, and border irregularities. CNN architectures, therefore, are able to capture these details without extensive manual feature engineering.

In our model, we designed a simple CNN architecture consisting of three convolutional blocks followed by two fully connected layers. Each convolutional block includes a convolutional layer with ReLU activation and a max-pooling layer, progressively increasing the number of feature maps (from 16 to 64) while reducing the spatial dimensions of the image. This allows the CNN to extract increasingly abstract and spatially invariant features.

The final feature maps are flattened and passed through a layer with 256 neurons and a dropout regularization layer to reduce overfitting. The output layer is a fully connected linear layer with the number of neurons equal to the number of skin disease classes, producing the class scores for classification. This architecture maintains a balance between complexity and computational efficiency, making it well-suited for image-based skin disease classification.

3.2 Baseline: Logistic Regression

We introduce a **logistic regression** to determine whether an image corresponds to a certain type of skin disease or not as well as attaching a probability value to it. This baseline model serves two purposes:

- Provides a straightforward model to measure improvement.
- Demonstrates if a more advanced multi-class approach (CNN) produces significant gains over a simpler classifier approach.

Upon testing this baseline model, it was evident that it performed poorly in terms of prediction accuracy. This can be due to the fact that logistic regression is too simplistic and cannot capture the many complex features. Moreover, it assumes that the relationships between image features are all linear, which is typically not true. These limitations could be further amplified by imbalances within the dataset.

3.3 Comparison with State-of-the-Art: Vision Transformer

Vision Transformers (ViTs) have recently shown promising results in computer vision tasks by leveraging self-attention mechanisms to model images as sequences of non-overlapping patches, similar to tokens in natural language processing. Unlike Convolutional Neural Networks, which focus on learning local patterns through convolutional kernels, ViTs are capable of capturing long-range dependencies across an entire image from the very first layer. ViTs use a linear projection of flattened patches to feed into the Transformer Encoder allowing for global relationships to be captured. This global receptive field is beneficial in dermatological applications, where subtle spatial relationships between such as asymmetry, irregular borders, or uneven color distribution may span across the whole image.

In this project, we include a pre-trained ViT model to evaluate whether this architecture can offer performance benefits over our baseline CNN. We fine-tuned the ViT on our skin lesion dataset to assess its generalization ability in a transfer learning setting. The Vision Transformer (ViT) significantly outperformed the Convolutional Neural Network (CNN), achieving a test accuracy of 99.16% compared to the CNN's 82.13%. This performance gap highlights the ViT's superior ability to generalize in the context of skin lesion classification. The ViT's near-perfect accuracy suggests

that it was better able to model the complex visual patterns of dermatological conditions, even in cases where lesions may appear visually similar. However, ViT's typically require large datasets to generalize well to the data. Due to our dataset not being as large as most datasets used for ViT's, it is also probable that the ViT is memorizing data and that the score it achieved is not representative of it's true generalization.

4 Evaluation

4.1 Performance Metrics

We will systematically track the following metrics during training and testing to evaluate model performance and ensure alignment with our clinical goals.

- **Accuracy (Target: $\geq$ 90%):** The percentage of correctly classified images. This is our primary baseline metric. We aim for at least 90% accuracy to ensure our model is usable for clinical contexts [1].
- **F1 Score (Target: $\geq$ 85% macro-average):** A macro-averaged F1 score of 85% or higher is our target, indicating reliable and fair performance across all classes, not just the most frequent ones. We defined 85% or higher because it is a common threshold for evaluating Machine Learning Models [5].
- **Precision and Recall (Class-wise Targets: $\geq$ 80%):** To reduce false positives and false negatives, we will monitor precision and recall for each class. For critical classes (e.g., melanoma), we aim for recall $\geq$ 90% to minimize the risk of missed diagnoses (false negatives). For all other classes, we aim for both precision and recall of at least 80% [6].
- **Area Under the Precision-Recall Curve (AUPRC, Target: $\geq$ 85% macro-average):** This is especially important for imbalanced datasets such as ours, where certain skin disease categories have far fewer samples. A macro-average AUPRC of 85% or higher will indicate that the model maintains strong discrimination capability across all classes [7].
- **Confusion Matrix Analysis:** In addition to the quantitative metrics, we will use the confusion matrix to identify specific patterns of misclassification. A key objective is to ensure that no high-risk class (e.g., malignant conditions) has a false negative rate above 10% [8].

By monitoring these metrics, we can fine-tune model hyperparameters and adjust the decision threshold to reflect our preference for fewer false negatives. For instance, if the model identifies a disease with low confidence, we might still flag it for follow-up rather than dismiss it completely.

4.2 Evaluation of our CNN model

When evaluating the CNN model, the following checks were performed to gauge how well it was performing when given a balanced dataset:

- **Data Augmentation Validation:** Validate how augmented samples improve F1 scores and reduce overfitting by comparing metrics with/without augmentation.
- **Generalization to Minor Classes:** Evaluate confusion matrices for each disease separately, checking that minority classes are not overlooked.

4.2.1 CNN with unscoped data

When evaluating the CNN on our unscoped data, the following metrics were obtained:

Test Accuracy	AUPRC	F1 Score
38.31%	0.1534	0.1478

Table 1: Performance metrics for CNN with unscoped data

As shown by the results above, the CNN performs extremely poorly when using this imbalanced

dataset, which is generally expected behaviour. The low test accuracy indicates that it struggles to differentiate between and accurately identify different skin diseases in the testing set. In addition, the low F1 and AUPRC scores indicate that there is low precision and recall due to the model classifying too many false positives and too few true positives. Furthermore, if we examine a heatmap that illustrates the distribution of classifications across the testing set, we see that the inaccurate classifications arises because the model is favoring the majority class. As shown in Figure 3, a

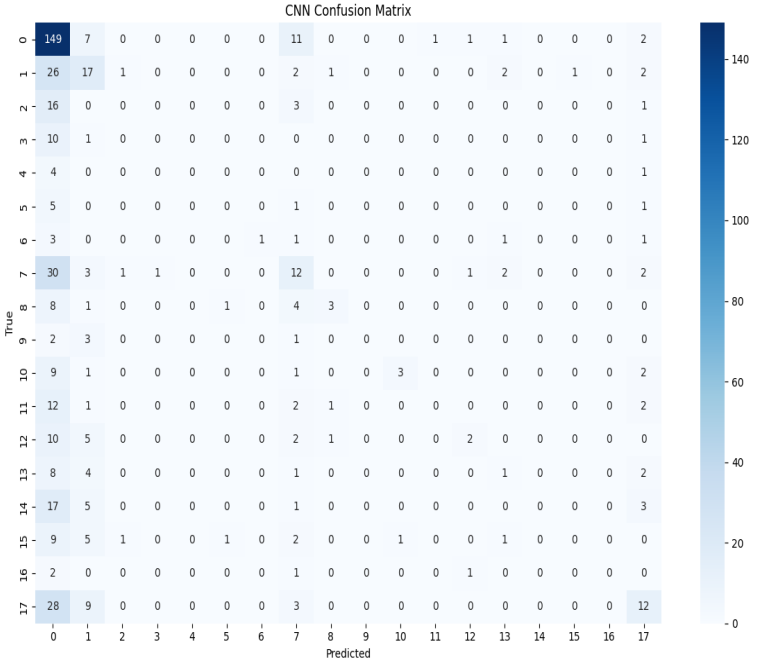


Figure 3: Confusion matrix of CNN with imbalanced dataset

majority of the models classifications are concentrated on the class with label 0, which is the majority class, thus demonstrating how this imbalance hinders the performance of our model.

4.2.2 CNN with scoped-down data

Upon scoping down the dataset, we can see an immediate improvement within our metrics as shown in the table below: The model improves in its ability to classify different skin diseases such as shown

Test Accuracy	AUPRC	F1 Score
68.48%	0.2135	0.1447

Table 2: Performance metrics for CNN with scoped data

by the increase in test accuracy. Moreover, the AUPRC score is improved as well due to the fact that a more balanced dataset is used. However, the F1 score is still rather low, indicating poor precision and recall.

4.2.3 CNN with scoped-down, balanced data

Upon using a balanced dataset augmented with synthetic data for the underrepresented classes, it is evident that the model performs significantly better in all metrics, as shown in the table below: The model performs much better when given a dataset that is balanced with synthetic data, as it has a higher test accuracy score as well as AUPRC and F1 scores, meaning there is a higher level of precision and recall. This is the result of balancing the dataset properly ensuring that there is not one specific class being heavily favoured over others. Looking at the performance of the model on this

Test Accuracy	AUPRC	F1 Score
82.13%	0.9026	0.8176

Table 3: Performance metrics for CNN with scoped, balanced data

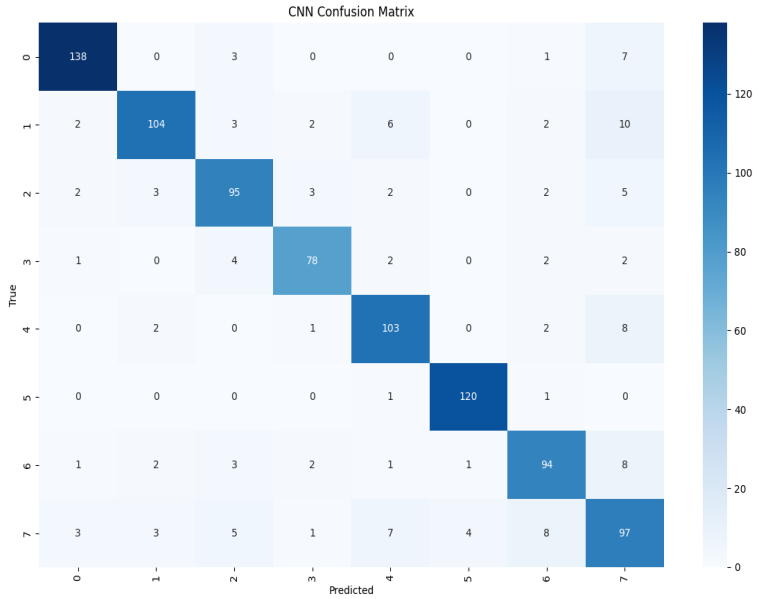


Figure 4: Confusion matrix of CNN with synthesized dataset

dataset based on the heatmap shown in Figure 4, it is evident that the CNN is generally effective at making correct classifications. The confusion matrix shows strong diagonal dominance, indicating that the majority of predictions align with the ground truth labels across most classes. For example, the model correctly classified 138 instances in class 0, 104 in class 1, and 120 in class 5, with only a few misclassifications.

However, there are still some notable off-diagonal values, indicating confusion between specific classes. For instance, class 1 was occasionally misclassified as class 7 (10 times), and class 7 showed minor confusion with several classes, particularly class 6 and class 0. Despite these cases, the overall distribution of misclassifications remains low relative to the number of correct predictions, which suggests that the model has learned to differentiate between classes effectively.

4.3 Comparison of Architectures

After training each model (baseline logistic regression, CNN, and Vision Transformer) under the same data splits, we will:

- Compare the final accuracy, F1 scores, and AUPRC across the three models.
- Analyze confusion matrices to see if one architecture systematically improves classification of certain diseases.
- Assess compute costs (training time, memory footprint) to analyze feasibility in typical healthcare deployments.

The **baseline model**, Logistic Regression Model, performed poorly with an accuracy of only 28.12%, a low AUPRC of 0.3406, and an F1 score of 0.2394, indicating both poor generalization and poor class balance handling. These metrics do not come close to meeting the conditions of success we set up earlier demonstrating that the Logistic Regression Model is too simplistic to capture complex features from image data.

The **CNN model** showed a large improvement from the baseline model, achieving an accuracy of 82.13%, an AUPRC of 0.9026, and an F1 score of 0.8176. This demonstrates that convolutional

architectures are well-suited for image pattern recognition in skin lesion classification, successfully balancing precision and recall even in potentially imbalanced datasets.

The **state-of-the-art ViT model** outperformed both the CNN and Logistic Regression models by a large margin, achieving an accuracy of 99.16%, AUPRC of 0.9970, and F1 score of 0.9928. This suggests that the Vision Transformer's ability to model long-range dependencies and global image features makes it particularly effective for diagnosing skin diseases. The ViT model not only classified most samples correctly, but did so with very few false positives or false negatives, meeting clinical safety standards and surpassing the 90% success threshold defined earlier.

	BASELINE	**CNN**	**STATE-OF-THE-ART**
Test Accuracy	28.12%	82.13%	99.16%
AUPRC	0.3406	0.9026	0.9970
F1 Score	0.2394	0.8176	0.9928

Table 4: Performance comparison of baseline, CNN, and state-of-the-art (ViT) models across key evaluation metrics.

5 Improvements

Here are the steps we can take to improve the project for the future:

- **Handle class imbalance:** Enhance the data loader by applying techniques such as:
  - Weight penalties (loss weighting)
  - Masking
  - Under-sampling or oversampling of minority/majority classes
  - Use more complex loss functions such as focal loss
- **Tune evaluation bias:** Adjust global metrics such as F1-score and precision to:
  - Bias the model toward false positives over false negatives
  - Simulate the cautious diagnostic approach a physician would take
- **Test on external datasets:** Evaluate the model using additional datasets or real-world clinical images to:
  - Validate generalizability across image resolution, lighting conditions, and demographics
  - Utilize other data augmentation techniques such as stretching or adjusting contrast or changing background colours to tune the model for more noisy inputs
  - Detect overfitting

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A Appendix / supplemental material

Implementation Details:

- **Data Splits and Cross-Validation:** We intend to use 5-fold cross-validation. Each fold holds out a unique 20% of the data for validation.
- **Preprocessing:** Images will be normalized and resized to a standardized size (e.g., 224×224 pixels).
- **Training Parameters:** We plan to use a cross-entropy loss function, stochastic gradient descent or Adam optimizer, and early stopping criteria to avoid overfitting.
- **Hyperparameters:** Learning rate, batch size, and number of epochs will be tuned. If time allows, we will conduct a grid search or use Bayesian optimization to find optimal values.

Implementation Tools:

- **Frameworks:** Likely PyTorch or TensorFlow/Keras. The code and data will all be stored on a GitHub repository at the following link: `https://github.com/manigel8/ECE324-Project.git`
- **Hardware:** GPU-based environment is expected to accelerate training.

Planned Experiments:

- Compare CNN, baseline logistic regression classifier, and Vision Transformer on the same subset of Dermnet data.
- Vary threshold settings for the logistic regression classifier to examine sensitivity to false negatives.
- Investigate how data augmentation and weighted loss functions impact F1 scores and confusion matrices.

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The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: We claim that we aim to achieve high accuracy and a focus on minimizing false negatives, and in the introduction we emphasize early detection and improved diagnostic support. These themes are shown throughout the modeling approach and conclusion.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

Justification: In Sections **??** and 5, we indicate that the Dermnet dataset can be imbalanced. We also mention that we only focus on a subset of diseases and prefer more false positives. These points clarify where the limitations of our approach might be inapplicable in a real clinical setting.

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   Answer: [NA]

   Justification: Our work is experimental and does not introduce theoretical models or proofs. We instead apply established deep learning methods in our machine learning approach (CNNs).

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5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We will be sharing a GitHub repository that contains all of our data, whether it is raw or processed, as well as any code that we use throughout the project.
The GitHub repository can be found at this link: `https://github.com/manigel8/ECE324-Project.git`
The link to our raw data can be found here: `https://www.kaggle.com/datasets/shubhamgoel27/dermnet`

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